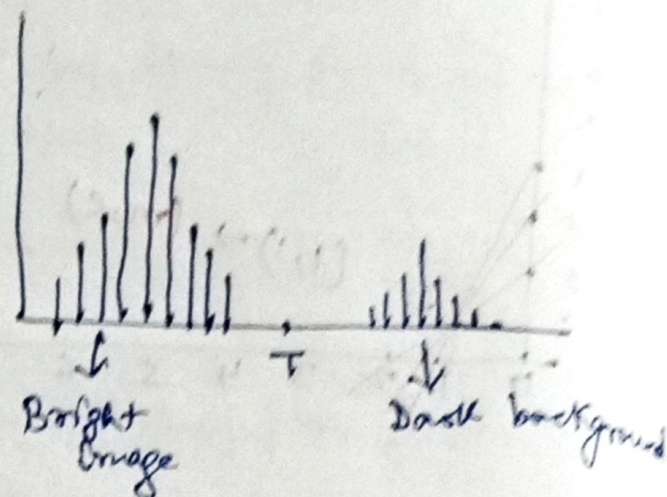
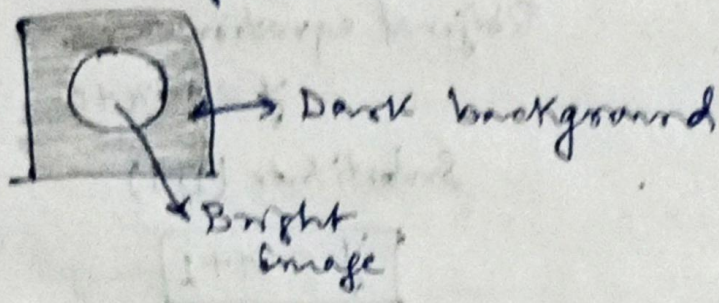


Thresholding

Thresholding $\rightarrow$ The assumption is that the range of intensities covered by objects of interest is different from the background.



Steps :-

- (a) A threshold T is selected.
- (b) Any point (x, y) in the image at which $f(x, y) > T$ is called an object point.
- (c) The segmented image denoted by $g(x, y)$ is given by

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

Types of Thresholding

- (1) Global thresholding - T is constant.
- (2) Variable thresholding - T changes over an image.
- (2) ~~or~~ Local thresholding - T is determined for localized regions within the image based on pixel intensity distribution within those regions.
- (3) Dynamic or adaptive thresholding - In variable thresholding, if the value of T depends on the spatial coordinates (x, y) .